

# MALI ANURAG H R

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## SUMMARY

Recent B.Tech graduate in Electronics and Communication Engineering with hands-on experience building and shipping AI/ML systems end-to-end — from classical ML and deep learning (LSTM, CNN/YOLO-based computer vision) to modern GenAI pipelines using LLM APIs, vector databases, and workflow automation. Comfortable across the ML lifecycle: data preprocessing, model integration, API design, and deployment. Seeking an AI/ML Engineer role to build production-grade, data-driven systems.

## TECHNICAL SKILLS

**Languages:** Python, SQL, C

**ML / Deep Learning:** TensorFlow, Scikit-learn, LSTM, OpenCV, YOLO, Predictive Modeling, Anomaly Detection

**GenAI / LLM Engineering:** Gemini API, Prompt-based batch inference, Vector Embeddings, Pinecone, n8n workflow automation

**Data:** Pandas, NumPy, Data Cleaning & Preprocessing, Power BI (DAX, Power Query)

**Frameworks & Tools:** Flask, Django, Git/GitHub, Excel

## PROJECTS

### AI News Aggregation & Voice Agent

*n8n, Gemini API, Pinecone (Vector DB)*

- Built an automated pipeline scraping 7 live RSS feeds (~376 articles/run) across national and world news, using n8n as the orchestration backbone.
- Designed batched LLM inference (10 articles/call via Gemini) to cut API calls from 376 to ~39 per run, avoiding rate limits.
- Engineered a custom dedup/cross-verification module that clusters articles by shared keywords, merges sources, and flags stories confirmed by 2+ outlets.
- Stood up a Pinecone vector index (3072-dim) for semantic retrieval, calling Google's embedding API directly after debugging a broken native integration; building toward voice delivery with follow-up Q&A (Groq for low-latency inference).

### AWHA — Anomaly Detection in Industrial IoT

*Python, Scikit-learn, LSTM*

- Built preprocessing and validation pipelines for multivariate industrial IoT time-series sensor data.
- Applied LSTM and Scikit-learn models to flag abnormal system behavior for downstream monitoring dashboards.

### AI-Powered Agricultural Assistant Chatbot

*TensorFlow, Flask*

- Built a plant disease and nutrient-deficiency classification system using a fine-tuned TensorFlow model.
- Integrated a Flask-based conversational interface with image processing pipelines for real-world agricultural use cases.

### Driver Drowsiness Detection System

*OpenCV, LSTM*

- Developed a real-time computer vision system to detect driver fatigue from live video streams.
- Implemented LSTM-based sequence detection with automated alerting to improve road safety.

### Warehouse Automation Robotic Arm

*YOLO, Arduino*

- Built a YOLO-based object detection and classification system for automated warehouse sorting.
- Integrated hardware-software communication, programming Arduino-based control for robotic arm movement.

## EDUCATION

**Amrita Vishwa Vidyapeetham, Bengaluru** — B.Tech, Electronics and Communication Engineering 2022 – 2026

**DR Academy** — Pre-University, State Board 89%

**Varin International Residential School** — Secondary School, ICSE 81.6%

## EXTRACURRICULAR ACTIVITIES

- Executive, Smriti — University Photography Club: organized workshops and cross-team collaborative media projects.
- Movies Division Lead, Dastaan Media Team: produced promotional content and coordinated live event media coverage.
- Core Committee Member, National Level Fest: managed vendor and team operations for a revenue-generating unit during a large-scale multi-day event.